

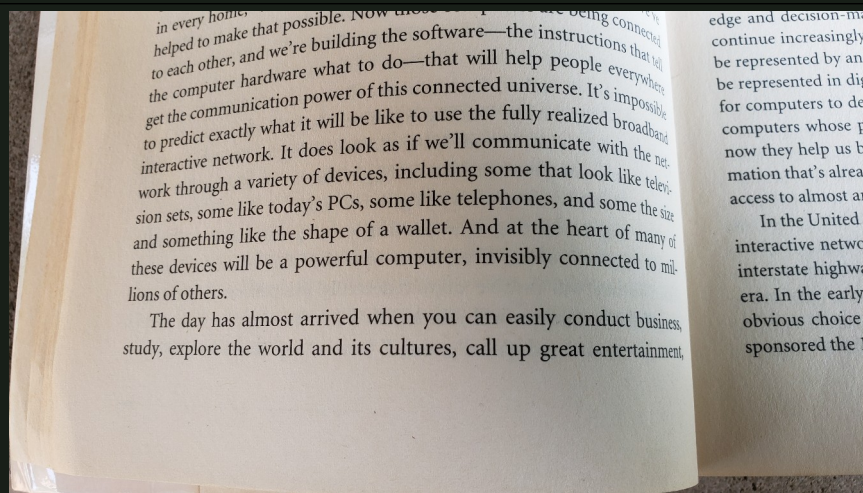
Ok, so I first read this book in June (2020), and I only read maybe the first 100-150 pages. The book is a bit difficult to follow in a lot of places, there's just a LOT of <technical information> in it that really only makes sense when you understand what he's talking about. Even though he carefully explains a lot of information, it is still a LOT of information to process.

I had to reread a bunch of what he said to make sense of it, but I managed to crush the whole thing in like a couple days once I got goin with it.

I decided to take some (pictures/notes) of the things he predicted. He essentially had like a <printing press of ideas> coming out of his head, and then he winds up wrapping everything up talking about building his house on <Lake Washington> with <Douglas Fir> trees.

"You don't want to use a <Douglas Fir> tree that's like (70) years old cause that's lame. Shit splits real easily. You need like dense-ass (500) year old <Douglas Fir> tree to REALLY build a house, dude."

Yeah, he didn't say that verbatim, but he makes plenty of humorous statements.



<20260723_081501.jpg>

Predicted Pocket/Wallet PC ⇒ Smartphones

So, he wasn't the first person to predict the idea of a smartphone, I believe <Gary Kildall> from <Digital Research> was MAYBE the first person to have that idea...? But I don't think it was really a unique idea, a lot of people probably thought of it.

literature. For the first time, people outside the canonical elite had access to written information.

The global interactive network will transform our culture as dramatically as Gutenberg's press did the Middle Ages.

Personal computers have already changed our work habits, but it is the evolving Internet that will really change our lives. As information machines are connected on the Internet, people, entertainment, and information services suddenly become accessible. As the Internet's popularity and capability increase, you'll be able to stay in touch with anyone, anywhere, who wants to stay in touch with you and to browse through any of thousands of sources of information, day or night. A little further along, you'll be able to answer your apartment intercom from your office or answer any mail from your home. Your misplaced or stolen camera will send you a message telling you exactly where it is, even if it's in a different city. Information that once was difficult to retrieve will be easier and easier to find:

Is my bus running on time?

<20260723_084730.jpg>

Predicted Surveillance cameras + Smartphone capabilities, GPS + E911

Being able to predict how "your misplaced or stolen camera will send you a message telling you exactly where it is, even if it's in a different city."

Yeah because <GPS> would be incorporated into the <pocket PC>.

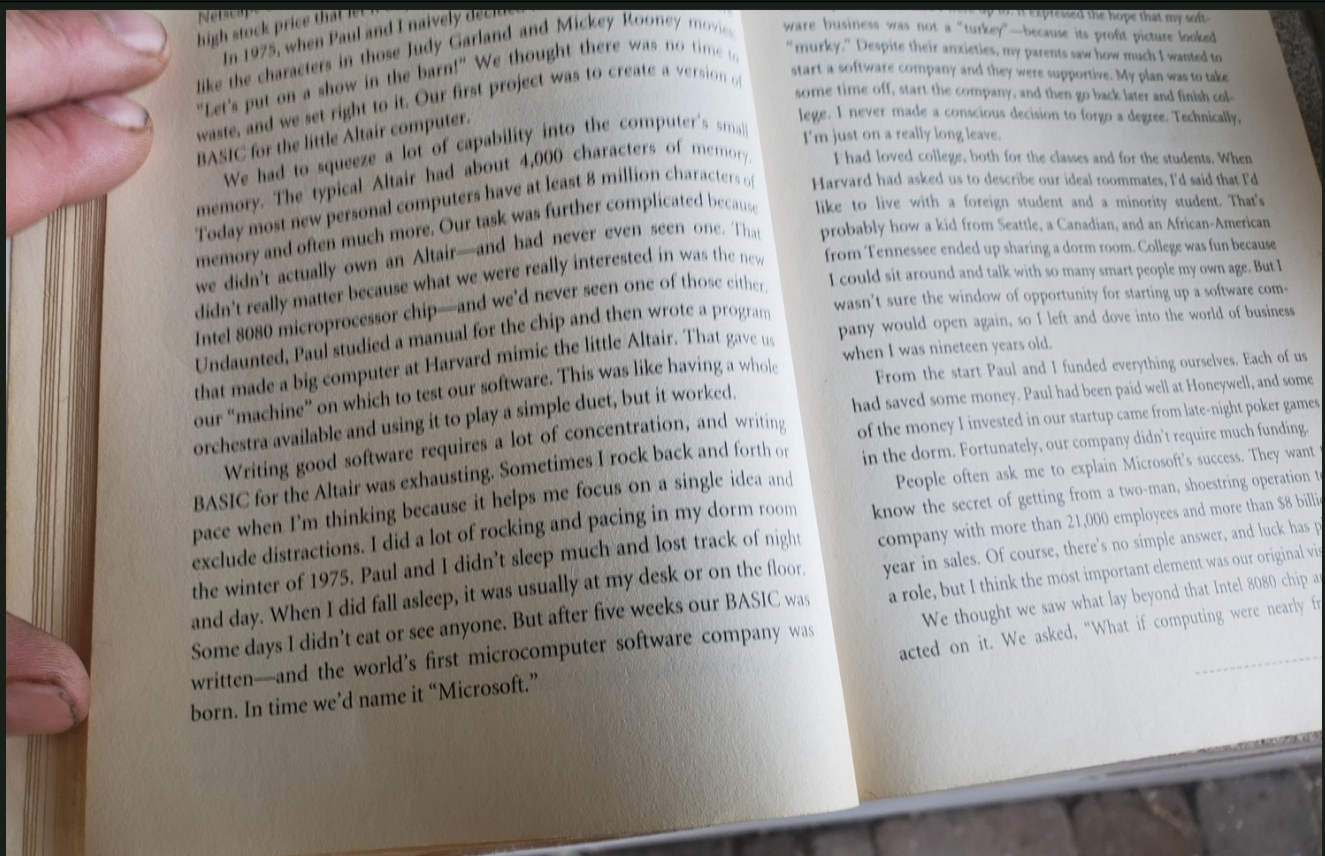
at that time, was a temperamental monster and a controlled cocoon. After we had used up the Mothers' Club grant, Paul Allen and I spent a lot of time trying to get access to computers. The computers performed modestly by today's standards, but they seemed awesome to us because they were big and complicated and cost as much as millions of dollars each. They were connected by phone lines to clackety Teletype terminals so that they could be shared by people at different locations. We rarely got close to the mainframes themselves.

Computer time was very expensive. When I was in high school, it cost about \$40 an hour to access a time-shared computer using a Teletype—for that \$40 an hour you got a slice of the computer's precious attention. This seems odd today, when some people own more than one PC and think nothing of leaving their computers idle for most of the day. Actually, it was possible even then to own your own computer. If you could afford \$18,000, Digital Equipment Corporation (DEC) would

<20260723_085652.jpg>

"Computer time was <very expensive> ... seems odd today when some people think nothing of leaving their computers <idle> for most of the day."

Yeah, I would imagine after having spent a <bunch of money> at the <computer lab> at <Lakeside school>, and the trouble they went through to build <BASIC>, that it would seem pretty odd that you would have a <device on standby> all day long.



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<Paul Allen> wrote (first version of?) BASIC for the <Altair 8800 + Intel 8080> using a <DEC PDP-8> to <compile the instructions>, <Gates + Allen> funded <everything themselves> and didn't ask parents for \$300K like <Jeff Bezos> did.

I can imagine how that conversation went...

Jeff : Hey mom, dad...

Jeff's Dad : Hey son.

Jeff's Mom : Hey son.

Jeff : Hey, can you guys loan me <300K> so I can <start a business> that will <lose money> for like, (20) years...?

Jeff's Dad : UH... what?

Jeff's Mom : ...<lose money> for (20) years?

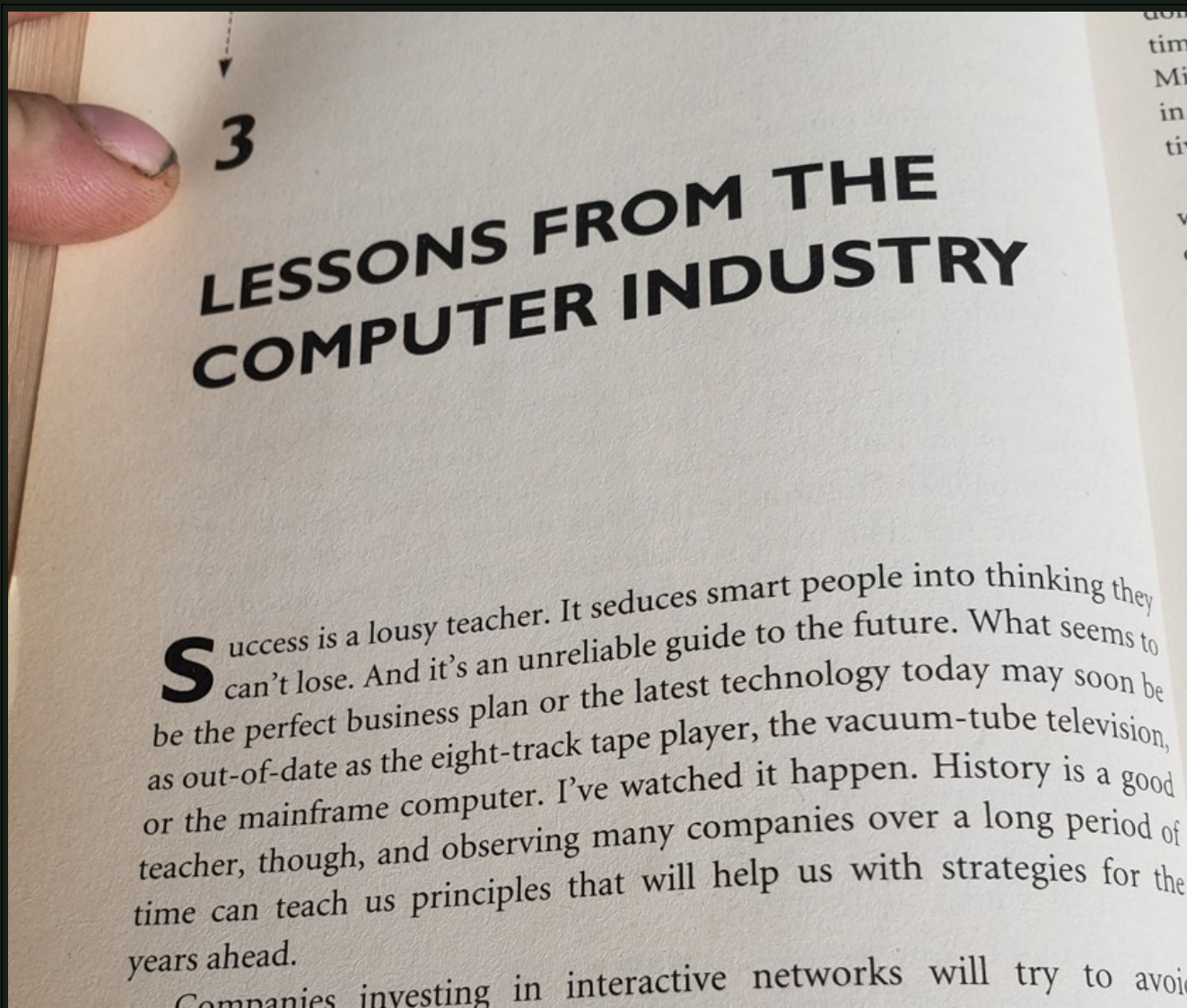
Jeff's Dad : That's a fuckin' stupid idea, son ...

Jeff : Ya, well ... You know what I always say, "Big things start small".

And then his father shook his head, but then they gave it to him ANYWAY so he could <lose money>, <recoup the lost money> with <rebates> from the <book publishers> for <volume of sales>, and ensure that <key investors> got a <return on their investment>. Now THAT'S smart.

The rest is <history>.

<Jeff> has done quite well for himself at <Amazon>, but was never really on <Bill's> level. You start the company with <0>, and then work your way up from there. That's a LOT harder to do, pal.



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Success is a <lousy teacher> → Failure is the <best teacher>.
Words I've lived by, but the <statements are different>.

Because, in life, in order to be <successful> and <REMAIN successful>, you have to be <willing to fail> more than the <competition> or <anybody else>, in order to become a <master of your domain>.

If you're NOT <willing to fail>, then <success will always feel rather fleeting>.

mind. In any case, I wasn't all that worried about it. I was having
In college a lot of posturing goes on, and looking as if you do
to study, as if you can slack off, can be considered a great way to
your coolness. During my freshman year I instituted a deliberate
of skipping most classes and then studying feverishly at the ex
term. It became a game—a not uncommon one—to see how
grade I could pull while investing the least time possible.

I filled a lot of my free time playing poker, which had its ov
tion. In poker, a player collects different pieces of informatio
betting boldly, what cards are showing, what this guy's pattern
and bluffing is—and then crunches all that data together to de
for his own hand. I got pretty good at this kind of inform
cessing. I played about three times a week. Games usually incl
dozen other students and lasted well into the night—or the
wasn't unusual for me to win or lose \$500 or \$600 a night
once I found myself thinking, “Oh God, it's eight o'clock in
and I've been sitting at this poker table for twelve hours an
posed to do my homework. Why did I stay in the game? I
and I feel awful. But if I play one more hand, maybe I'll h
\$300.” It wasn't always fun.

The poker strategizing experience would prove helpf
into business, though. The other game I played at coll
poning one, wouldn't serve me well at all. I didn't know t
In fact, I was encouraged in my dilatory practices beca

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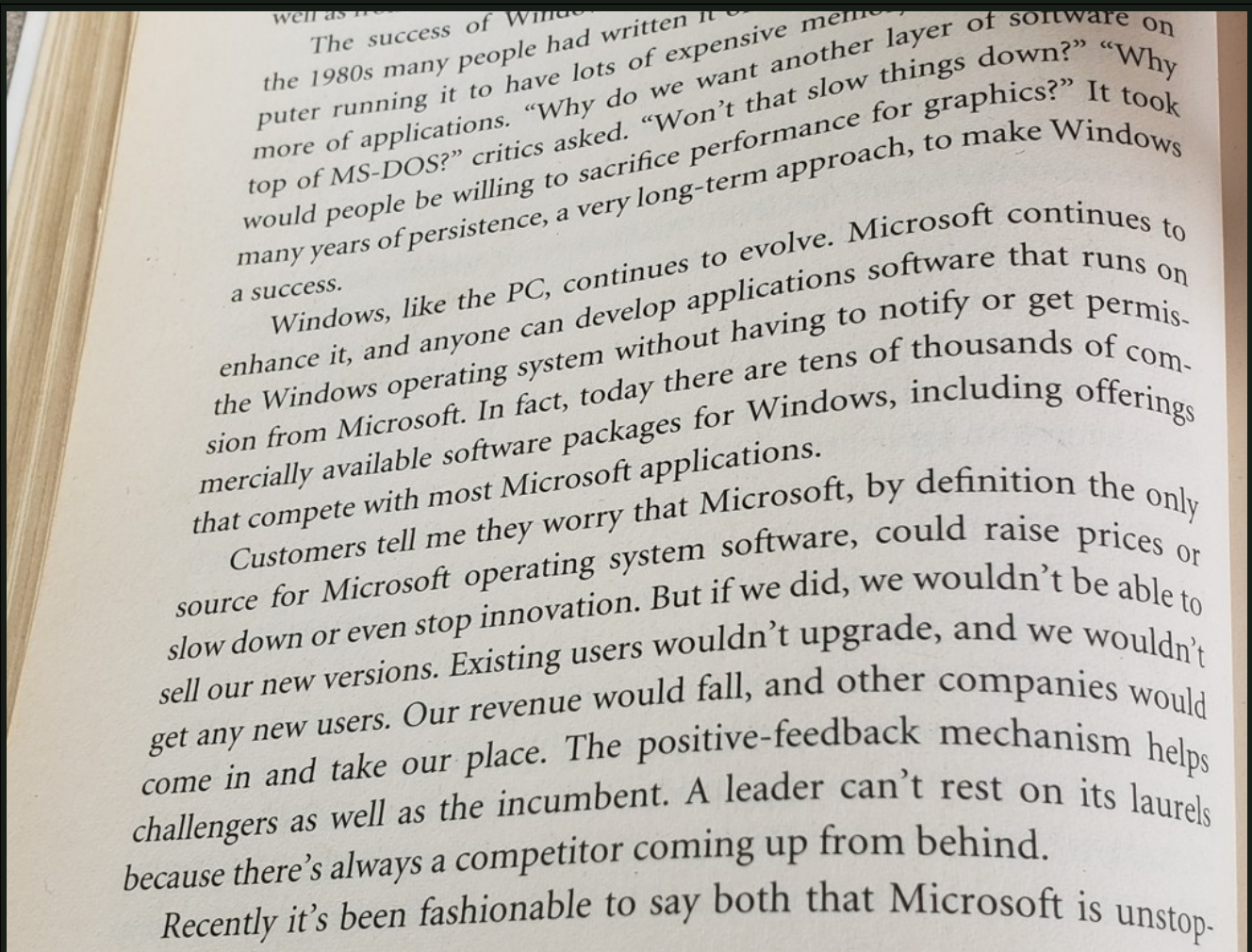
Dude made a game out of <attending the fewest number of classes> so he could go kick

people's asses at <poker>, and would <cram study session> the night before an exam with <Steve Ballmer> (I think?), in order to <see how high of a grade he could get>.

<Least possible effort> → <maximum possible result>.
Only people who KNOW they're smart will do this.

He mentions that this <postponing game> wouldn't serve him well at all...
Nah, <I think it was his best strategy that whole time>.

If the dude wants to spend hours playing <Minesweeper>...
Let him do it, for fucks sake.



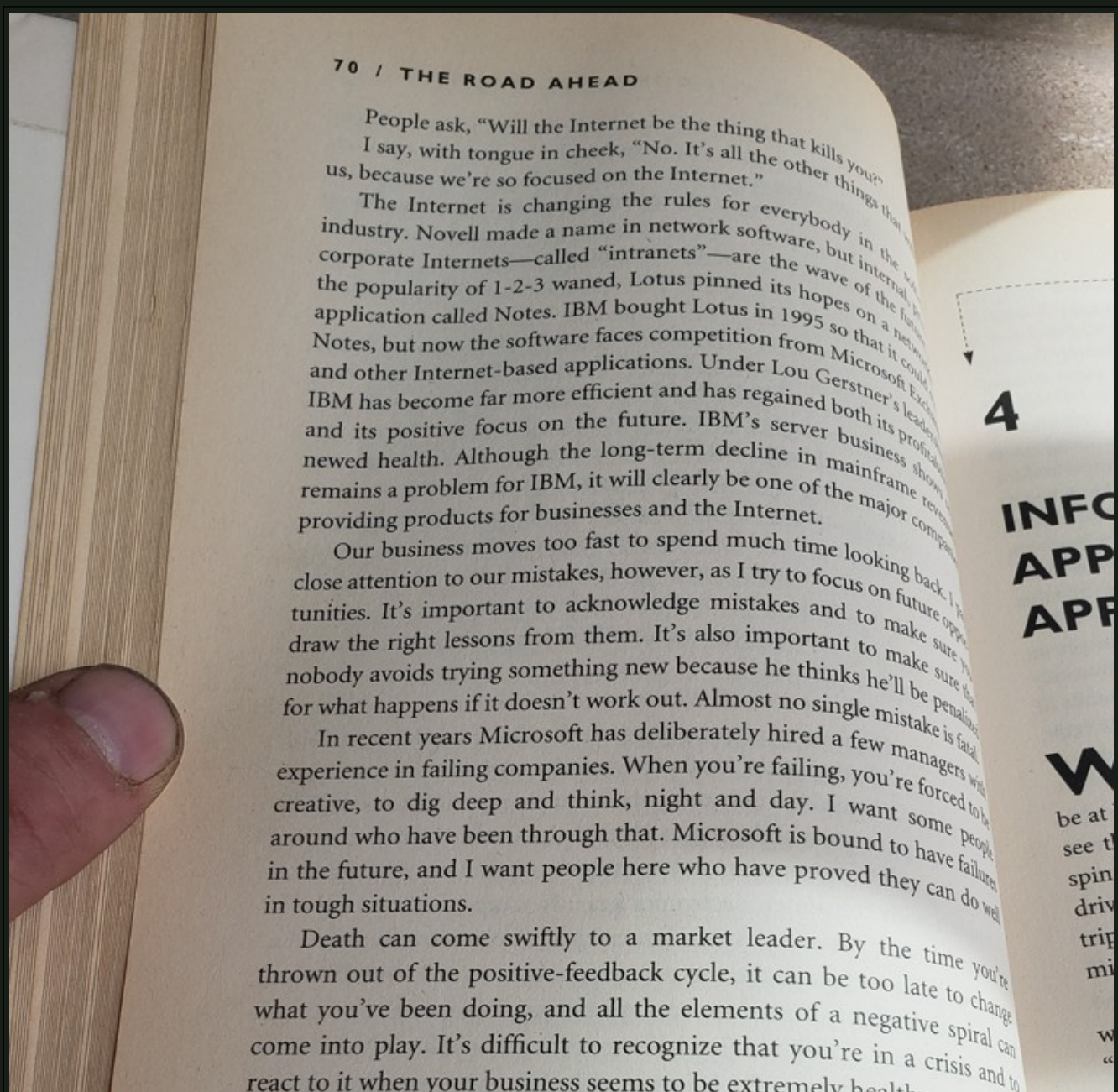
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Success of <Windows> took a long time to (shape/mold), and faced a lot of <competition>.

<OS/2> collaboration with <IBM> was a <fucking nightmare>, and <scared Mr. Gates>, because they wound up investing a lot of <time + resources> into helping build <OS/2> only for them to wind up using the work they did to compete against <Microsoft>.

However, that plan <backfired>.

This is what you would call a <backstabbing move>. It definitely <backfired on IBM> for choosing to turn their backs on a <co-developer>.



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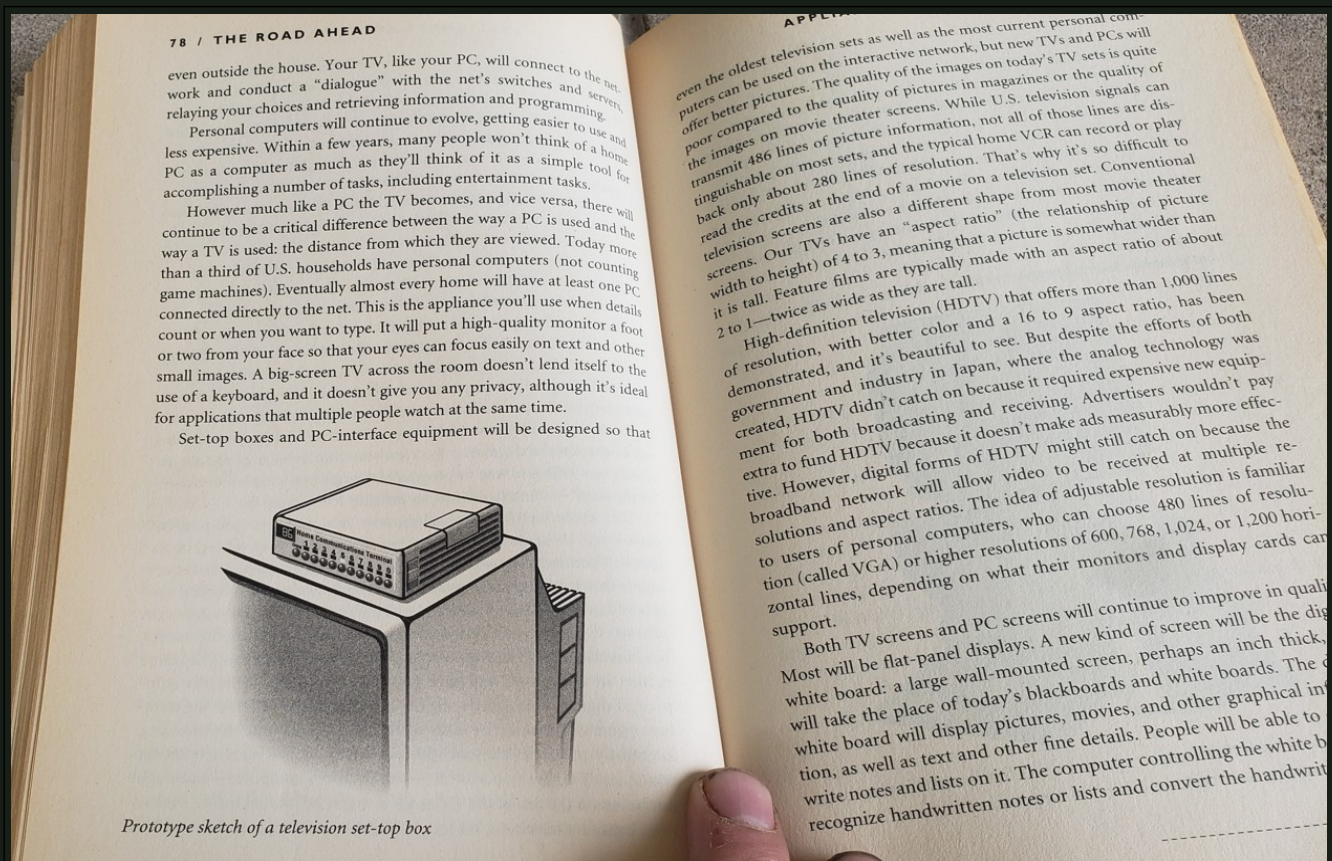
Dude has a good sense of humor.

People ask 'Will the internet be the thing that kills you?'

'Nah, its all the <OTHER> things that will kill us, because were so <focused on the internet>.'

Continues to talk highly about <IBM> even AFTER they stabbed him in the back.

Not sure how the hell he kept his cool that whole time. Dude either has the <patience of a nun>, or just does a <really good job hiding his frustrations>.



<20260726_112826.jpg>

Predicted that <Smart TV>'s, <desktop/laptops>, and <smartphones> would be <separate devices>.

Talks about <display resolution> and <VGA/HDTV>, <16:9 aspect ratio>, improvements in <visual quality>, etc. Knew that the <early internet> was <limited>, and could not provide a <rich experience> until <ADSL/ISDN> were providing <mid-band experience>.

But even THAT was <unsuitable> for <high definition video>. Wouldn't be until actual <broadband> hit the scene, where that could begin to <flourish on the internet>.

Or, allow the <internet to flourish>.

OR, the internet would only flourish once some <REAL major telecom shit> started to go down <across the country>.

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they're on the road. Two centuries ago a traveler could carry a "lap desk," a hinged writing board attached to a thin mahogany box with a drawer for pens and ink. When folded, it was reasonably compact, and when opened, it provided an ample writing surface. *The Declaration of Independence* was written on a lap desk in Philadelphia, a long way from Thomas Jefferson's Virginia home. The need for a portable writing station is met today by the laptop, a folding, lap-size personal computer. Many people who work from both office and home—including me—choose a laptop (or a slightly smaller computer, known as a notebook) as their primary computer. In the office, the small computer can be connected to a large monitor and to the corporate network. Notebook computers will continue to get thinner until they're about the size of a tablet of paper.

Notebooks are the smallest and most portable real computers today, but we'll soon have pocket-size computers with snapshot-size color screens. When you whip one out, nobody will say, "Wow! You've got a computer!"

What do you carry on your person now? Probably at least keys, identification, money, and a watch. And maybe credit cards, a checkbook, traveler's checks, an address book, an appointment book, a notepad, something to read, a camera, a pocket tape recorder, a cellular phone, a pager, concert tickets, a map, a compass, a calculator, an electronic entry card, photographs, and maybe a loud whistle to call for help.

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History → the <Declaration of Independence> was written on lap desk in <Philadelphia>, and THAT form made its way into <laptops>.. mentions preference for <laptop devices> + <docking stations>. (so do I)

Rattles off (20) devices that inevitably became (1) (Wallet PC/Pocket PC/smartphone).

Loud whistle to call for help → E911.

Says "We'll have pocket-size computers with snapshot size color screens. When you whip one out, nobody will say 'WOW DUDE, YOU'VE GOT A COMPUTER~! I'M BLOWN AWAY~!'"

Yeh, he was right, nobody says that LOL.

They don't even say:

"Is that a computer in your pocket, or are you just happy to see me?!?"

easy for anyone to spend and accept digital funds. Your wallet will link into a store's computer to allow money to be transferred without any physical exchange at a cash register. Digital cash will be used in interpersonal transactions too. If your son needs money, you might slip five bucks from your wallet PC into his digitally.

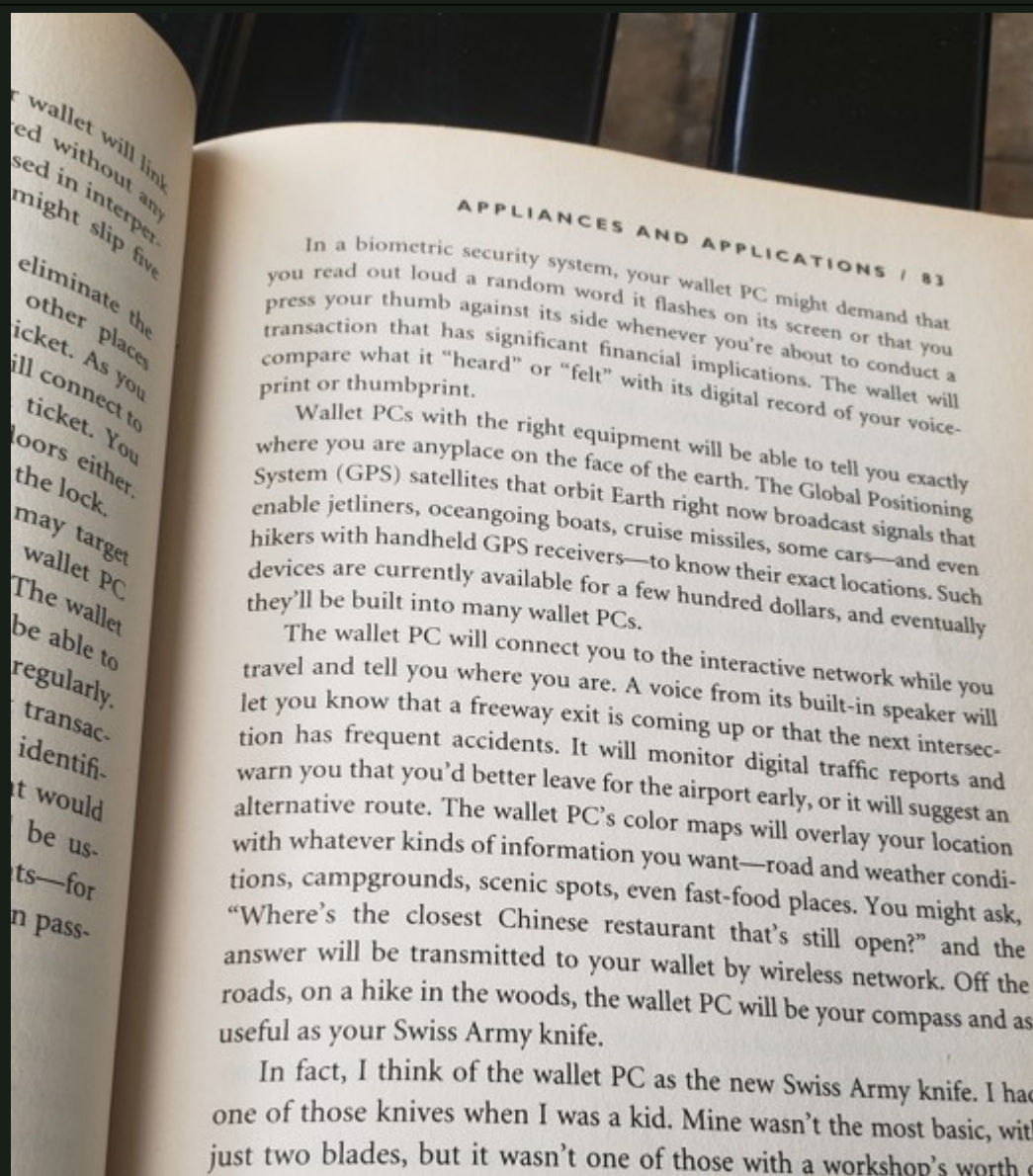
When wallet PCs have become ubiquitous, we can eliminate the bottlenecks that plague airport terminals, theaters, and other places where people queue up to show their identification or a ticket. As you pass through an airport gate, for example, your wallet PC will connect to the airport's computers and verify that you've paid for a ticket. You won't need a key or a magnetic card key to get through doors. Your wallet PC will identify you to the computer controlling the lock.

As cash and credit cards begin to disappear, criminals may turn to wallet PCs, so there will have to be safeguards to prevent a wallet PC from being used in the same way a stolen credit card can be. The wallet PC will store the "keys" you'll use to identify yourself. You'll be able to invalidate your keys easily, and you'll be able to change them.

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Predicts 1) <PayPal>, 2) <Zelle>, 3) <CashApp>, (electronic funds transfer? EFT is the banking side, but I think these programs all utilize EFT), 4) <online banking>, 5) <smart cards>, 6) <modern credit cards>, 7) <personal identification numbers>, etc.

<Dude obviously saw the future>.



<20260726_113920.jpg>

Predicts 1) <biometrics> used for <security>, 2) <multi-factor authentication>, 3) <GPS> in a <wallet PC> as useful as <Swiss Army Knife>.

<Wallet PC> AKA <Pocket PC> AKA <Smartphones> turned out to be FAR more useful than a <Swiss Army Knife>, actually... and <I don't say that lightly>.

Might be <taking liberties> in saying he predicted ALL of these things, but he <clearly mentions precursors> to ALL of this.

The browsing experience is surprisingly rewarding, although it's nothing like being in Paris.

If you're using spatial navigation, the place you're moving around in doesn't have to be real. Eventually you'll be able to set up imaginary places and return to them whenever you want to. In your own museum, you'll be able to move walls, add imaginary galleries, and rearrange the art. You might want all still lifes to be displayed together, even if one is a fragment of a Pompeian fresco that hangs in a gallery of ancient Roman art and one is a Cubist Picasso from a twentieth-century gallery. You'll be able to play curator and gather images of your favorite artworks from around the world to "hang" in a gallery of your own. Suppose you want to include a painting you remember with great affection of a man asleep being nuzzled by a lion. If you can't remember either the artist or where you saw it, you'll be able to describe what you want by posing a query. The query will start your computer or other information appliance sifting through reservoirs of information to deliver the piece that matches your request.

You'll be able to give your friends tours of your gallery, regardless of whether they're sitting next to you or watching from the other side of the world. "Here, between the Raphael and the Modigliani," you might say, "is a favorite finger painting I did when I was three years old."

The last type of navigational aid, and in many ways the most useful of all, is an agent—a query agent. It's a program that can find what you want for you. It's convenient for them to work, but they don't provide the future, age. An agent v remember yo work more e mation appl activities an. The first format a d was forma know abo becomes you like t. The explicit tant for its expe softer. experi So

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Predicts 1) <virtual reality>, 2) <augmented reality>, and talks about <specific art pieces> like (Pompeian fresco/ancient Roman art), <Picasso>, <Raphael>, <Modigliani>

(I don't know much about art, I'm sure <Richard Sackler> will think "Wow, dude doesn't know art? Wow, what a tool.")

convenient for people who use them often enough to remember how they work, but for the average user the graphical user interface by itself doesn't provide enough guidance to using the features. In the systems of the future, agents will remedy that.

An agent will know how to help you partly because the computer will remember your past activities. It will note patterns of use that will help it work more effectively with you. Through the magic of software, information appliances connected to the net will seem to learn from your activities and will make suggestions to you. I call this "softer software."

The first day a new assistant is on the job, you can't simply ask him to format a document the way another memo you wrote a few weeks ago was formatted. You can't say, "Send a copy to everybody who should know about this." But over the course of months and years, the assistant becomes more valuable as he picks up on what is routine and on how you like things to be done.

The computer today is like a perpetual first-day assistant. It needs explicit first-day instructions all the time, remaining a first-day assistant forever. It will never make one iota of adjustment in response to its experience with you. At Microsoft, we're working to make software softer. No one should be stuck with an assistant who doesn't learn from experience.

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Predicts 1) (Agentic AI/ChatGPT) (27) full-cycle earth-years before it existed, <agents/helpful assistants>, etc. (some were leap years)

Ya there's nothin' quite like ACCURATELY describing what eventually happened, but like, <decades> ahead of schedule. That's sayin' somethin'. I mean, maybe other people ALSO made these claims, but what he's sayin' here is <totally spot on>.

That's what they built. Maybe <Sam Altman> read this book too, and was like:

Altman : Hey, you know what?

Someone : Uh... what...?

Altman : Gonna build that shit <Bill Gates> talked about in <The Road Ahead>.

Someone : Yeah...?

Altman : Yeh.

Someone : Cool, dude.

And that was it.

evenly by any number except 1 and itself. Among the first dozen numbers, the primes are 2, 3, 5, 7, and 11. The numbers 4, 6, 8, 10, and 12 are not prime because 2 divides into each of them evenly. The number 9 is not prime because 3 divides into it evenly. There are an infinite number of prime numbers, and there is no known pattern to them except that they are prime. When you multiply two prime numbers together, you get a number that can be divided evenly only by those same two primes. For example, only the primes 5 and 7 can be divided evenly into 35. Finding the primes is called "factoring" the number. If we factor 35 we get the 5 and 7.

It's easy to multiply the prime numbers 11,927 and 20,903 and get the product 249,310,081, but it's much harder to recover from the product the two prime numbers that are its factors. This one-way function, multiplying prime numbers that are difficult to factor out of the product, underlies an ingenious and sophisticated encryption system. It takes a long time for even the biggest computers to factor a really large product back into its constituent primes. A coding system based on factoring uses two different decoding keys, one to encipher a message and a different but related one to decipher it. With only the enciphering key, it's easy to encode a message, but deciphering it within any practical period of time is nearly impossible. Deciphering requires a separate key, available only to the intended recipient of the message—or rather, to the recipient's computer. The enciphering key is based on the product of two huge prime numbers, whereas the deciphering key is based on the primes themselves. A computer can generate a new pair of unique keys in a flash because it's

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<Prime numbers>, <prime numbers>, <prime numbers>.
Talks about <prime numbers> and why they're <important to security>.

Gonna say it again, cause I noticed it a while back...
<Prime numbers> are important to <security>, but they also have a <logical equivalent> when <writing code> that isn't <redundant>.

You start with a <prime block>, and then you keep <adding> to it.

You can <reuse plenty of the code you write>, but the <classwork> is essentially a <prime number> in a way, or in some <abstract sense>.

numbers to generate an enciphering key, which will be listed publicly and a corresponding deciphering key, which only the user will know. This is how it will work in practice: I have information I want to send you. My information appliance/computer system looks up your public key and uses it to encrypt the information before sending it. No one can read the message, even though your key is public knowledge. You receive the message, and your computer decrypts it with a private key that corresponds to your public key.

You want to answer. Your computer looks up my public key and uses it to encrypt your reply. No one else can read the message, even though it was encrypted with a key that's totally public. Only I can read it because only I have the private deciphering key. This is a very practical encryption system because no one has to trade keys in advance.

How big do the prime numbers and their products have to be to ensure an effective one-way function?

Public key encryption was invented by Whitfield Diffie and Martin Hellman in 1977. Another group of computer scientists, Ron Rivest, Adi Shamir, and Leonard Adelman, soon came up with the notion of using prime factorization as part of what is now known as the RSA cryptosystem, after the initials of their last names. They projected that it would take millions of years to factor a 130-digit number that was the product of two primes, regardless of how much computing power was brought to bear on finding the solution. To prove the point, they challenged the world to find the two factors in this 129-digit number, known to people in the field as RSA 129:

114,381,625,757,888,867,669,235,779,976,146,612,010,218,296,
721,242,362,562,561,842,935,706,935,245,733,897,830,597,123,
563,958,705,058,989,075,147,599,290,026,879,543,541

They were sure that a message they had encrypted using this number as the public key would be totally secure forever. But they hadn't anticipated either the full effects of Moore's Law, as discussed in chapter 2, which has made computers much more powerful, or the success of the personal computer, which has dramatically increased the number of computers and computer users in the world. In 1993 a group of more than 600 academics and hobbyists from around the world began a

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(Diffie/Hellman) and <RSA>, took about (16) years for the industry to <crack their algorithm>, one that was using a 129-digit number. Which is why 256-bit AES is rather commonplace today. But that probably won't last long now with <quantum computing>.

The whole section is <super important> for ANYONE looking to take <security engineering> seriously.

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methodical assault on the 129-digit number, using the Internet to coordinate the work of various computers. In less than a year they factored the number into two primes, one 64 digits long and the other 65:

3,490,529,510,847,650,949,147,849,619,903,898,133,417,764,638,
493,387,843,990,820,577

and

32,769,132,993,266,709,549,961,988,190,834,461,413,177,642,
967,992,942,539,798,288,533

The encoded message says, "The magic words are squeamish and ossifrage."

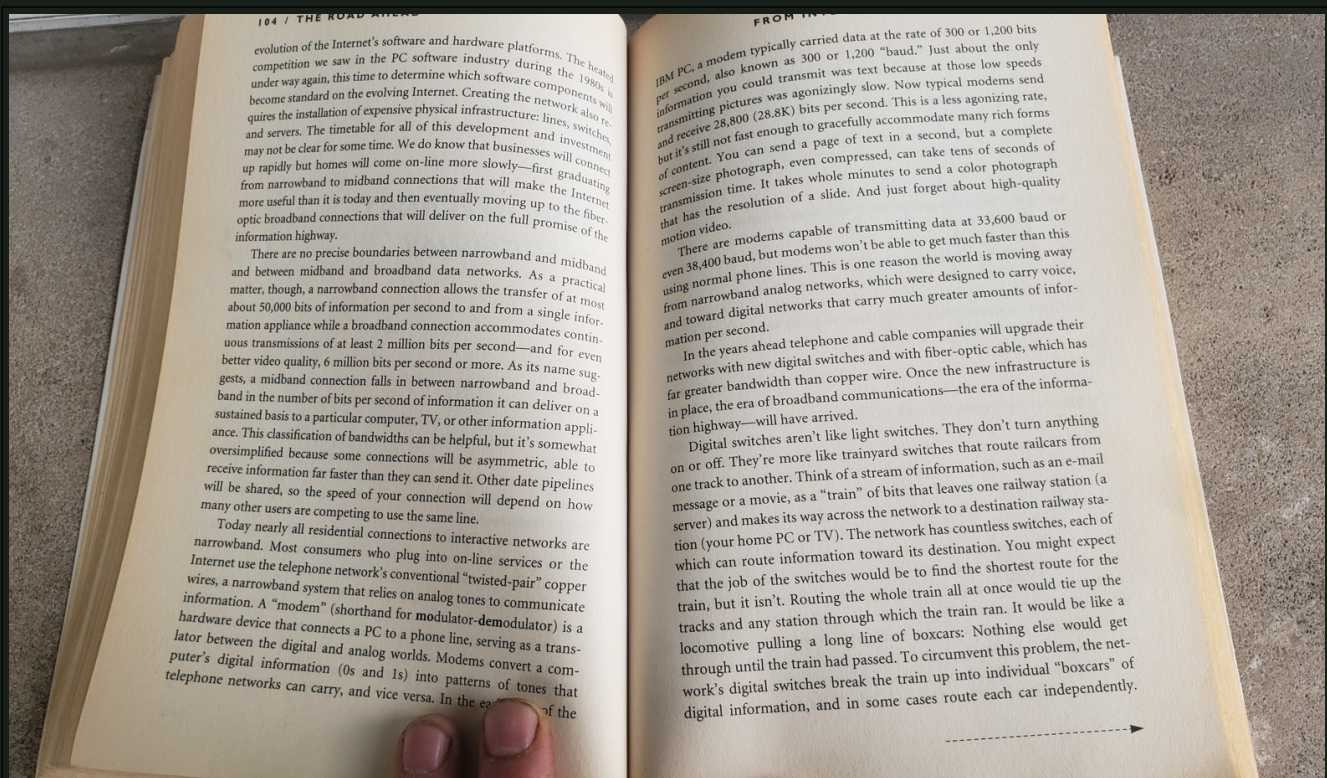
One lesson that came out of this challenge is that a 129-digit public key is not long enough if the information being encrypted is really important and sensitive. Another is that no one should get too cocksure about the security of encryption.

Increasing the key just a few digits makes it much more difficult to crack. Mathematicians today believe that a 250-digit-long product of two primes would take millions of years to factor with any foreseeable amount of future computing power. But who really knows? This uncertainty—and the unlikely but conceivable possibility that someone could come up with an easy way of factoring big numbers—means that networks will have to be designed in ways that allow encryption schemes to be changed readily.

The importance of this need for flexibility was demonstrated late in 1995 when a cryptography consultant, Paul Kocher, was able to break implementations of public key encryption without brute-force

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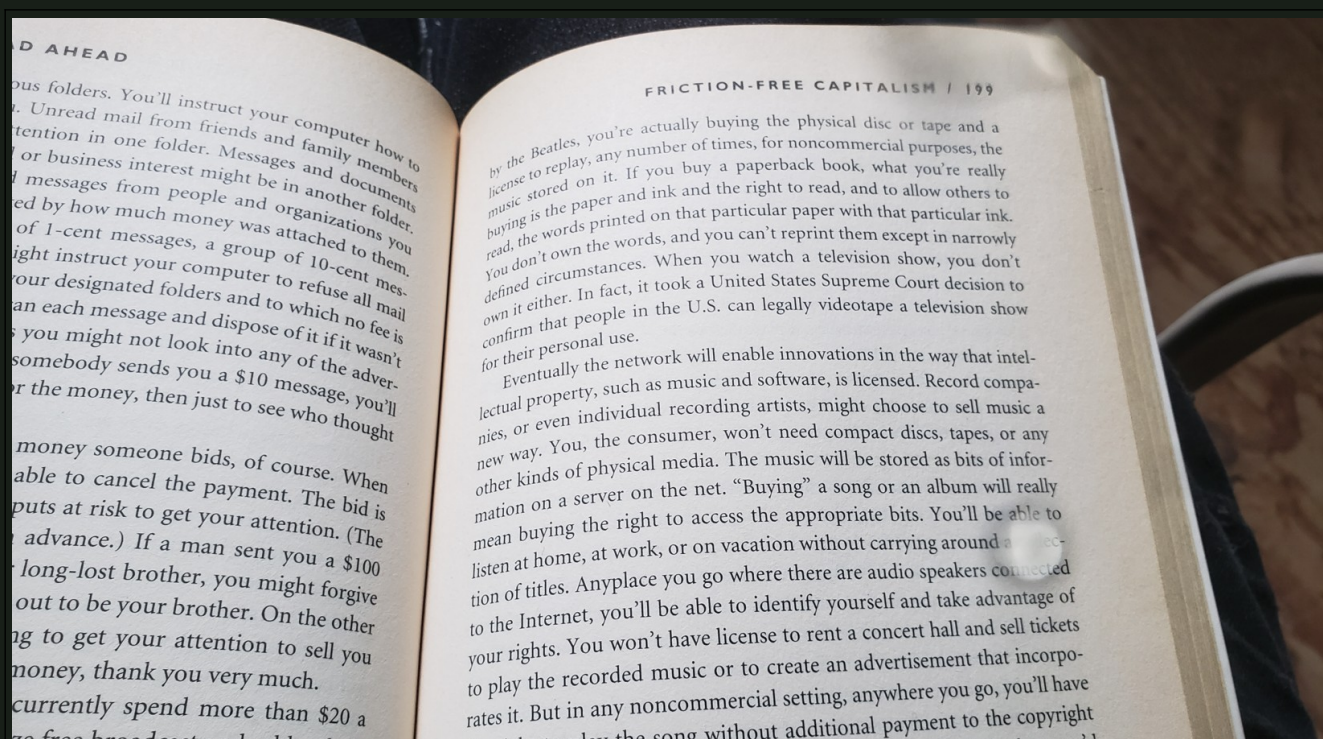
"Fortunately, it's easy to <modify decryption software> so that its' work cant be accurately timed. Unfortunately, there may be OTHER ways of <breaking public key encryption> that nobody's thought of <yet>. So there's always <something> to worry about." Uh, yeh. +1.



<20260728_072019.jpg>

Discusses <narrow/mid/broadband> internet connectivity and spends an entire chapter tiptoeing around not saying <modems suck ass> at <transferring pictures + video>. He eventually spends a fair amount of time discussing <fiber optics>.

But yeh, that's effectively what he's sayin', <modems sucked even back then>.



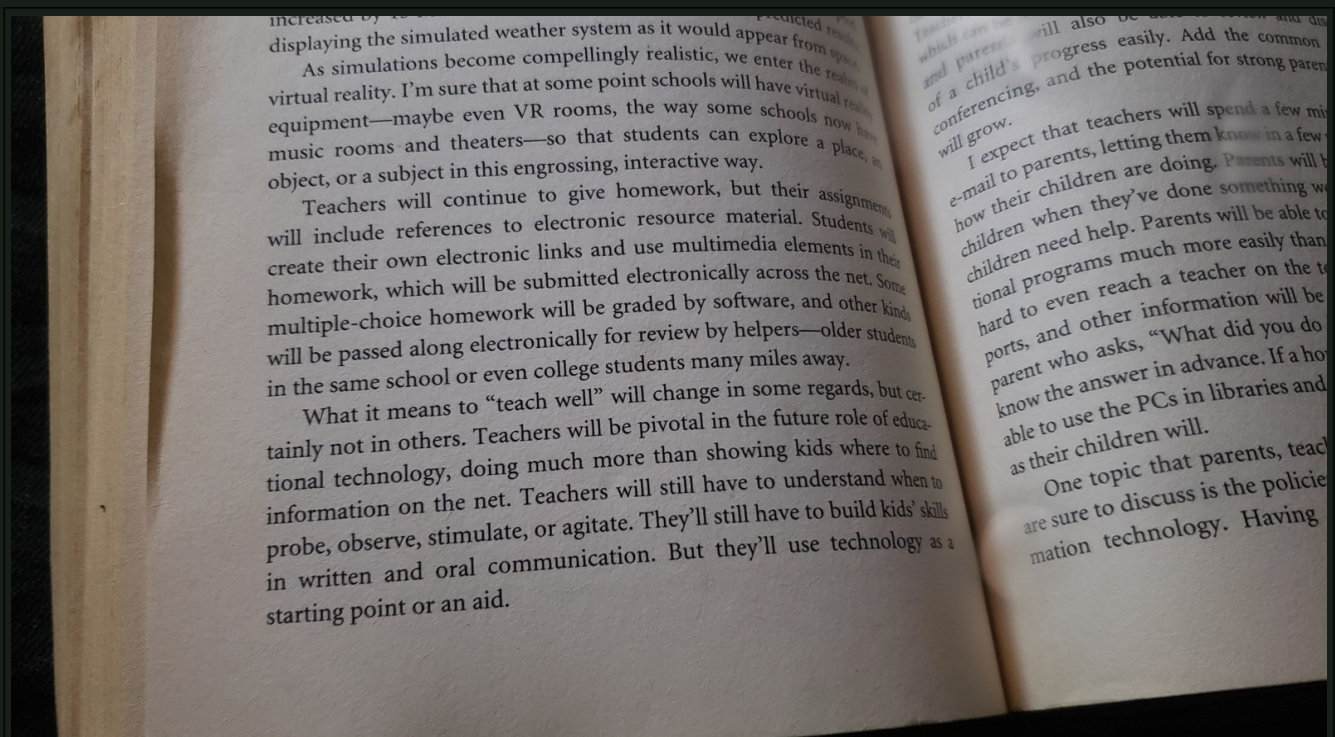
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Talks about how <copyrights> work. You buy a book, but you only own that <copy>. You can't <reprint it>, except in <narrowly defined circumstances> (such as these screenshots and my commentary = valid use), but also understands (where + how) certain forms of media would be more difficult to enforce copyright laws for, such as (sharing, music, <YouTube>, <Spotify>, whatever, etc.)

The same rules apply to the <Microsoft Windows Operating System>, you do not OWN a copy, you own a LICENSE to USE it. It is <still THEIR intellectual property>, and they reserve the right to <monitor the device> through the <Customer Experience Improvement Program/CEIP> and <Windows Update>...

...AND, to assist in the hunt for <terrorists> + <nation state actors> like <Advanced Persistent Threat 29> AKA <Cozy Bear>. This would not be made apparent to users unless they read the terms of service during the installation of <Windows>, and even then a lot of this is <written between the lines>.

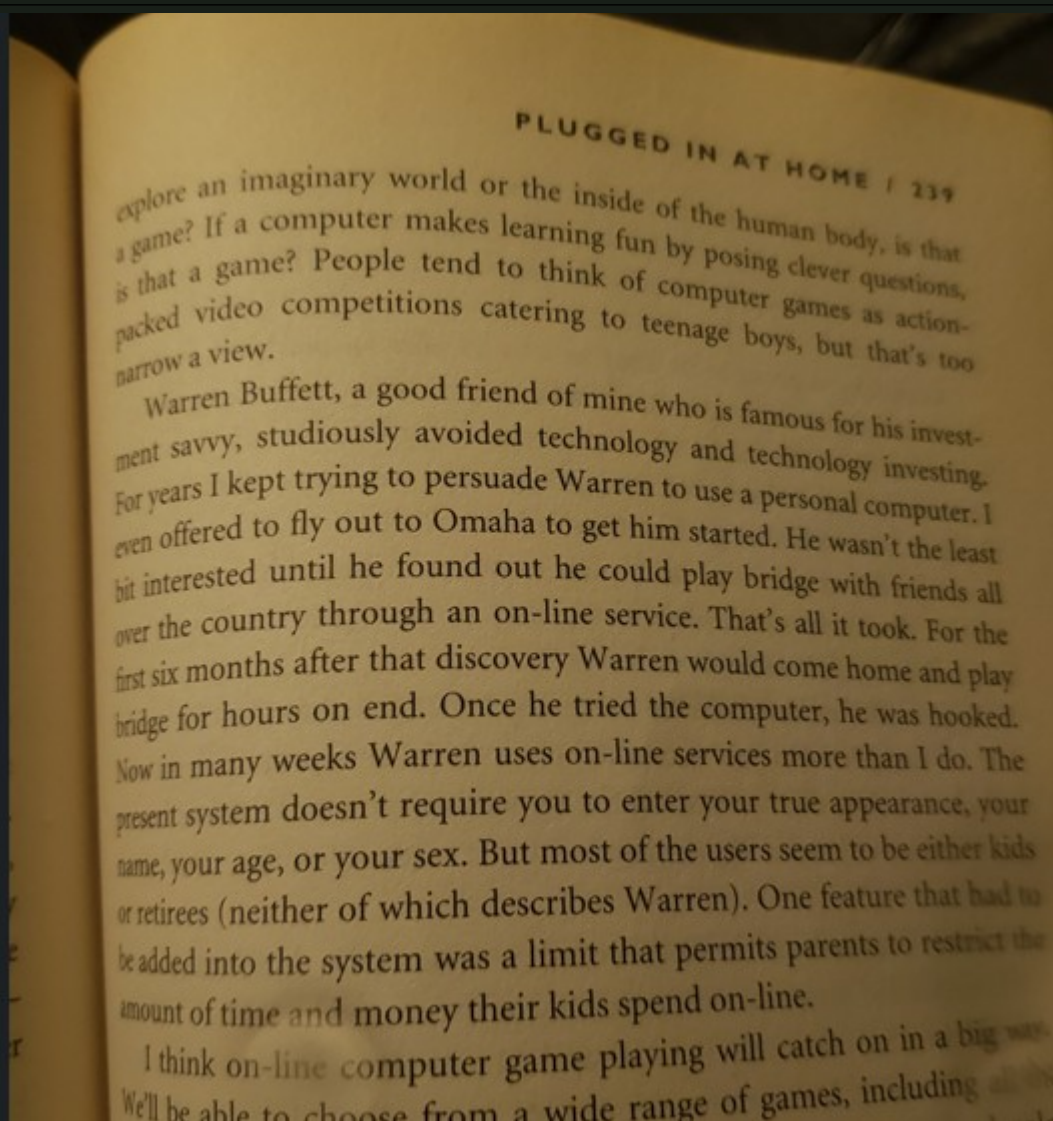
That means, if you're smart enough you can <decipher> what it is ACTUALLY saying.



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Teachers will still have to understand when to:
1) probe, 2) observe, 3) stimulate and 4) AGITATE.

Gee, I wonder what THAT means... maybe it means (when/where) to give a <student> a swift kick in the ass to keep them <motivated> for their own benefit.



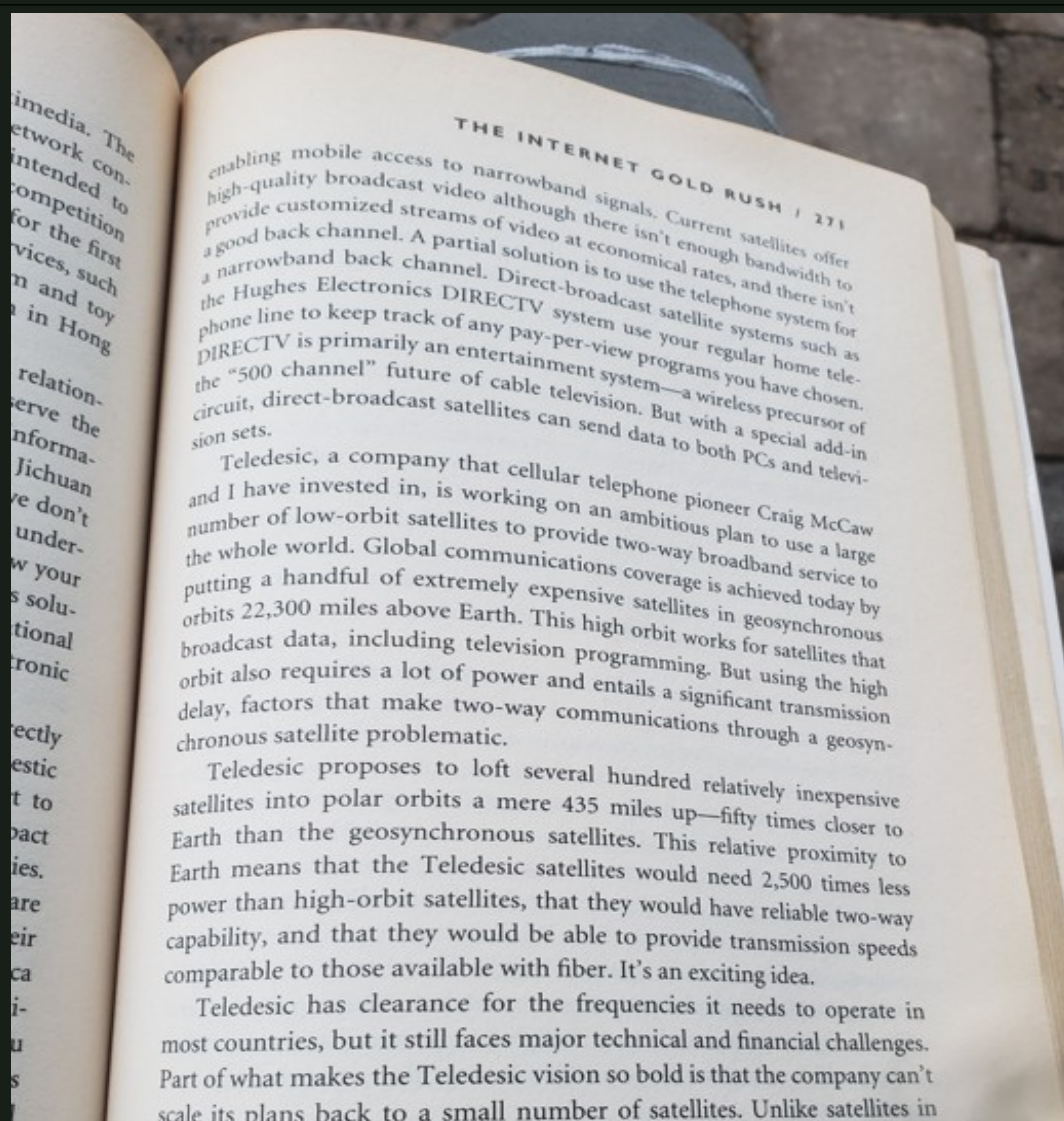
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Talks about his friend <Warren Buffett>, but <he and Warren> aren't friends anymore, cause of the <Epstein file> bullshit.

Wasn't much of a friend then. Sounds like a (posturing/optics) move on his part. Maybe I'm taking liberties by saying that... but-

I'd rather find out that he had <extramarital affairs> with <hot Russian blonde girls> (who probably looked a lot like <Ksenia>) than finding out he's a pedophile. Which I don't think he is, nor ever has been.

Anyone reading this, you'd probably fuck <Ksenia> too if you had the chance, even if you're a girl.



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Talks about <satellite connections> + predicted <Starlink> when the idea was still called <Teledesic> (30) years ago.

Wonder if <Elon Musk> became a <trillionaire> because HE read this book too...

Musk : Hey Sam.

Altman : Hey Elon.

Musk : Hey, remember when you said we should make a <nonprofit organization> dedicated to <open source AI>?

Altman : Yep.

Musk : Yeah well, how about YOU build the AI, and turn <OpenAI> into a <for-profit> organization so that I can later sue you in court and lose...?

Altman : Uh.

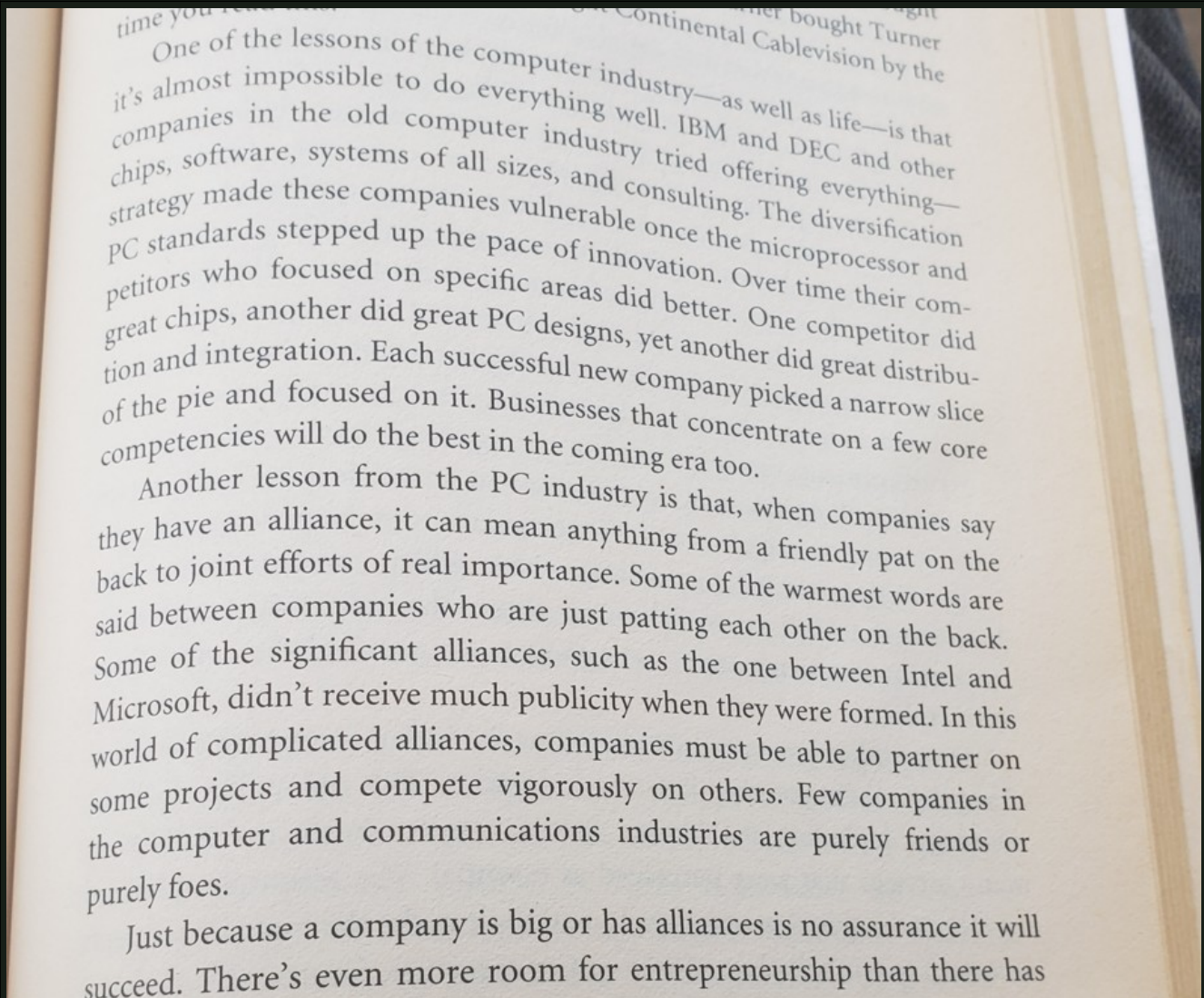
Musk : Just throwin' out ideas.

I really just wanna build <electric cars> and <rockets>, and make <Teledesic> a thing we could make <billions of dollars> off of.

Altman : Idk bro, fine, do your thing.

And that was that. Trucks and bulldozers.

Admittedly, <Tesla> and <SpaceX> are doing extraordinarily well, but that <inauguration day nazi-salute> DEFINITELY upset me. Dude did that to see how much he could get away with, and I won't stand for it.



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"When companies say they have an alliance, it can mean anything from a friendly pat on the back to joint efforts of real importance."

Recognizes the concept of <posturing> but in a business relationship, not at Harvard. <Posturing> is a lot of people just patting each other on the back, while CLAIMING that they did a <GREAT> job, not unlike what <Donald J. Trump> likes to do.

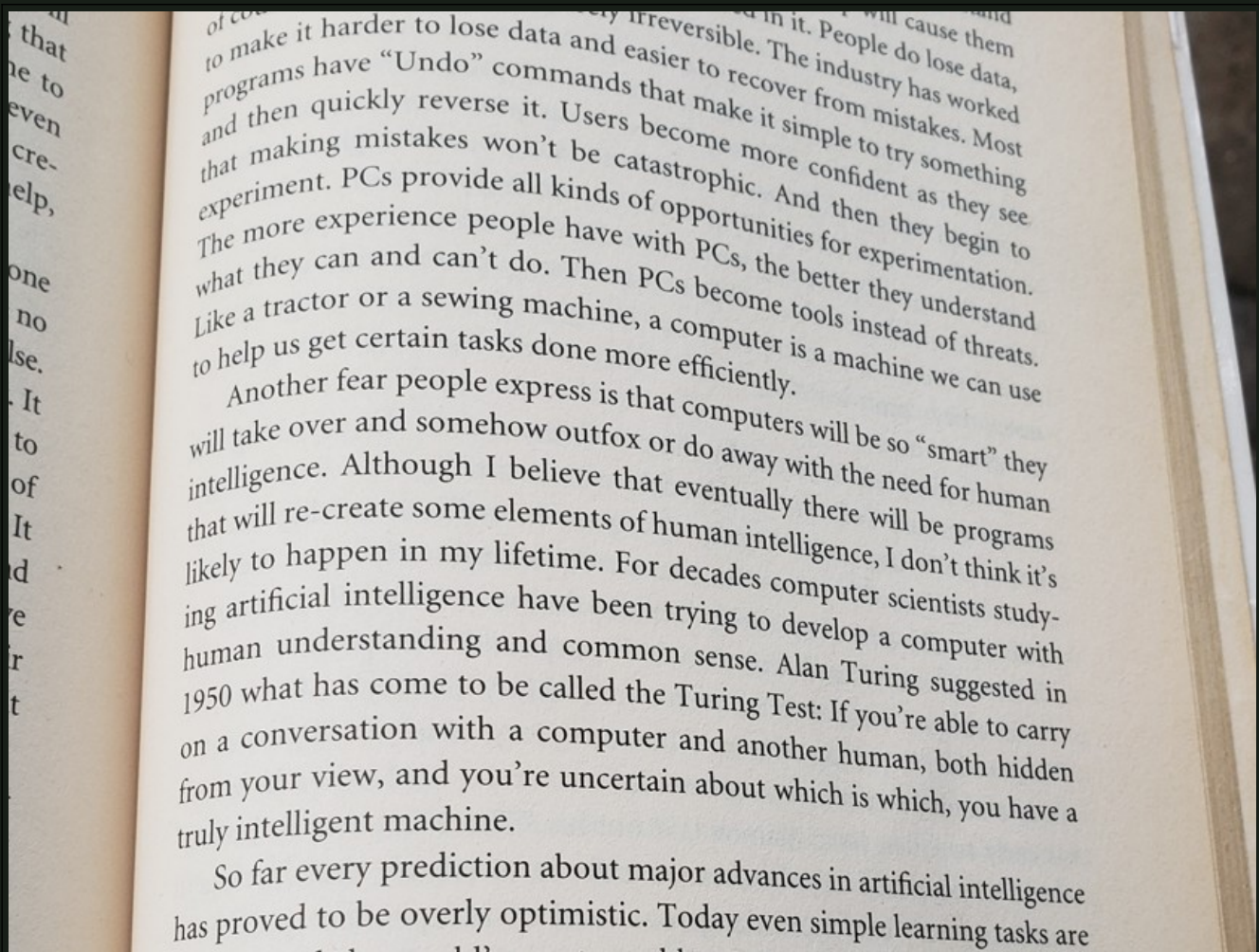
Ya, <Make America Great Again>, my ass. MAGA = <posturing>.

Also mentions how <Intel> and <Microsoft> were basically together in the trenches, crawling under barbed wire in the mud while being shot at way deep in enemy territory... (not verbatim)

That's how strong <Microsoft> + <Intel>'s relationship has always been, they were both the vehicle to each others success, and <Intel> never bailed on them like <IBM> did with <OS/2>.

At least not to my knowledge...

Maybe <Warren Buffett> should take someone else's advice for a change.



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Predicts AI being able to do WAY cool shit like beat the best <chess player> in the world (Gary Kasparov) via <Deep Blue>, talks about the <Turing test>, where you talk to a human, and you talk to an AI, and if you can't tell them apart, then it has <passed the test>.

Thing is, they still haven't built that. Like, maybe what they built passes the Turing test with SOME people, but someone like me is gonna notice pretty quickly.

Also, AI tends to OVERSATURATE people with <information + details>.

I can do that, apparently <Bill Gates> was also able to do that, so its not an inherent indication that WE are AI... but it COULD mean that the AI was modeled after the way WE think.

This isn't <JayzTwoCents>... it's MY (2) cents.

Support <JayzTwoCents>, he's cool in my book. <Skunkworks FTW edition>.

290 / THE ROAD AHEAD

World chess champion Gary Kasparov beat Deep Blue in a 1996 match, but even Kasparov admitted he had a scare when the computer won the first game of the series. That was the only game the computer won, though. Kasparov came back with three wins and two draws to win the match. The contest was made out to be a "human versus machine" showdown, but in a real sense the contest was between a human—Kasparov—and an opposing committee of programmers and chess players who used a tool—Deep Blue—to carry out their carefully predetermined strategies. Kasparov won that match, but he might not always win. When a machine finally prevails convincingly in chess, it shouldn't offend human dignity any more than the realization that a person with binoculars can see faraway objects better than a person without binoculars can. As long as computers are being programmed for specialized tasks, they aren't learning—and if they aren't learning, they pose no threat to humans.

I am troubled, however, by the possibility that in the very long run computers and software could achieve true intelligence. Predicting a timeframe is impossible because progress in artificial intelligence research is so incredibly slow. But once machines can really learn, they will be able to take over most things humans do today. This will raise issues of who is in control and what the whole purpose of our species is.

Of course, these worries aren't new. Mary Shelley's 1818 novel, *Frankenstein*, is perhaps the most famous story of an intelligent man-made creature that comes to control its creator. In 1942, when he was

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Chess continued, <Isaac Asimov: Three laws of robotics>, and <James Cameron: Terminator>, also discussed by <Computerphile> and various others, also worries about <empowerment of terrorists>...

Cause terrorists are fuckin' (stupid/crazy), man. They're willin' to blow themselves up in the name of <Allah>. What sorcery is THAT shit...?

too much for our own good. This could become a problem when virtual reality technology gets really good and becomes commonplace.

One day a virtual reality game will let you enter a virtual bar and make eye contact with "someone special," who will note your interest and come over to engage you in conversation. You'll talk, impressing this new friend with your charm and wit. Perhaps the two of you will decide, then and there, to go to Paris. Whoosh! You'll be in Paris, gazing together at the stained glass windows of Notre Dame. "Have you ever ridden the Star Ferry in Hong Kong?" you might ask your friend. Whoosh! VR will certainly be more engrossing than video games have ever been, and more addictive.

If you found yourself escaping into such an attractive world too often, or for too long, and you started to worry about it, you could impose some discipline on yourself by telling the system, "No matter what password I try, don't let me play any more than half an hour of games a day." This would be a little speed bump, a warning to slow your involvement with something you've found a little too appealing. It would serve the same purpose as a photo of some very overweight people you might post on your refrigerator to keep you from snacking.

Speed bumps help a lot with behavior that tends to generate day-after regrets. If somebody wants to spend his or her free time window gazing in a simulation of Notre Dame, or visiting with a synthetic friend in a make-believe bar, that person is exercising his or her freedom. Today a lot of people spend several hours a day with a television on. To

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Predicts VR like <Second Life> and all of its' derivatives, <time management with an agent>, self-imposed <speed bumps + discipline>, predicted how IT would (change/constitute) society's new playground, workplace + classroom.

our boom box. This versatility will be the strength of the network, but it will also mean we'll become extraordinarily reliant on it.

That much dependence can be dangerous. During the New York City blackouts in 1965 and 1977, millions of people were in trouble—at least for a few hours—because of their dependence on electricity. They counted on electric power for light, heat, transport, and security. When electricity failed, people were trapped in elevators, traffic lights stopped working, and electric water pumps quit.

A complete failure of the network is worth worrying about. Because the system will be thoroughly decentralized, any single outage is unlikely to have a widespread effect. If an individual server fails, it will be replaced and its data restored. But the system could be susceptible to assault. As the network becomes more essential to us, we'll have to design in more redundancy.

One area of vulnerability is the system's reliance on cryptography—the mathematical locks that keep information safe. None of the protection systems that exist today, whether steering-wheel locks or steel vaults, are completely fail-safe. The best we can do is make it as difficult as possible for somebody to break a security device or get inside. Despite popular impressions to the contrary, computer security has a very good record. Computers are capable of protecting information in such a way that even the smartest hackers can't get at it readily unless somebody who is entrusted with the information makes a mistake or is corrupt. Sloppiness is the main cause of computer security breaches. On the net—

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History, NYC dependence on <electricity>, <blackouts>, and why systems needed <decentralization> for success or <persistence> of uptime, and <CRYPTOGRAPHY>.

Computers are capable of <protecting information> in such a way that even the smartest hackers can't get at it readily unless somebody who is <entrusted with the information> makes a <MISTAKE> or is <CORRUPT>. <SLOPPINESS> IS THE MAIN CAUSE OF COMPUTER SECURITY BREACHES***. ← So, he's saying people need to <write code people can understand>.

AKA, predicted <ransomware>, <digital money/heists>, and <Pachinko scam> type shit.

use to pay for pachinko games—and which can be redeemed indirectly for cash.

Because both the Internet's privacy and the security of digital money depend on encryption, a breakthrough in mathematics or computer science that defeats the cryptographic system could be a disaster. It was sobering to me when certain kinds of public key encryption were revealed to be vulnerable to a "timing attack," a development I discussed in chapter 4. The obvious mathematical breakthrough that would defeat our public key encryption would be the development of an easy way to factor large numbers. Any person or organization who had this capability could counterfeit money, penetrate any personal, corporate, or government file, and possibly even undermine the security of nations, which is why we have to be so careful in designing the network's encryption system. We have to ensure that if any particular encryption technique proves fallible, there is a way to make an immediate transition to an alternative technique. There's a little bit of inventing still to be done before we have that perfected. The implications of Moore's Law make it particularly hard to guarantee security for information you want to be kept confidential for a decade or more. More powerful computers can lead to the breakthroughs that can violate our security.

Loss of privacy is another major worry where the network is con-

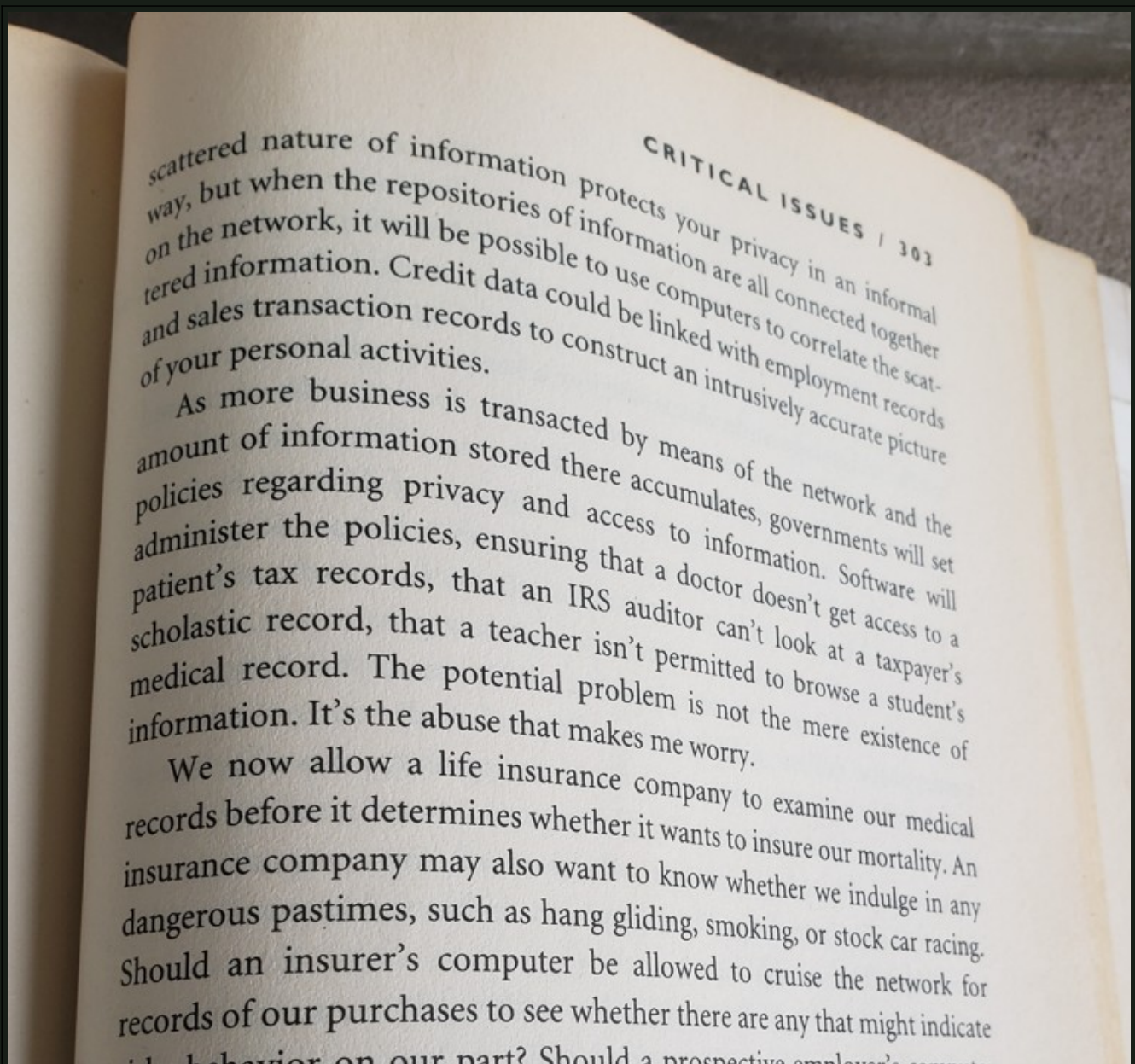
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Makes obvious statement about <TIMING ATTACKS>, and how <computer science + cryptography> + <smart dipshits> could <circumvent> the most <well-built systems>.

<Task scheduler/Panther> is often used to inject <delayed attacks>. <Timed-trigger attack>. Can lay <dormant> for <X amount of time>, and then executes without leaving much of a trace behind. (Except in forensic-level event logs like <SECURITY>)

However, using such an attack is a <UNIQUE IDENTIFIER> that can be <traced back> to an <advanced persistent threat>'s activities.

Also talks about how <companies> would <target users> with <advertisements> based on <data analytics>. Predicted <Cambridge Analytica> + other similar scandals.



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Discusses how everyone would have data about them ranging from <educational institutions>, <health/medical institutions>, <tax records>, and <partitioning that information> so it wouldn't ALL be available to <whoever> or be <misused>, even though it is, and continues to be <abused/misused>.

I've learned first-hand that a lot of people don't actually give a shit about <HIPAA> or <PCI compliance> violations, they care about <posturing> and <optics>, that's it.

I could expand on this observation I've made, if asked.

...will be the ultimate diary and autobiography, if you want one. If nothing else, you'd know exactly when and where you took a photograph when you organized your family's digital photo album.

The technology to do this is not just a remote possibility. It should soon be possible to compress the human voice down to a few thousand bits of digital information per second, which means that an hour of conversation will be converted into about 1 megabyte of digital data. Small tapes used for backing up computer hard disks already store 10 gigabytes or more of data, enough to record about 10,000 hours of compressed audio—more than a year's worth. Tapes for new generations of digital VCRs will hold more than 100 gigabytes, which means that a single tape costing a few dollars could hold recordings of all the conversations an individual has over the course of a decade or possibly even a lifetime—depending on how talkative he is. These numbers are based on today's capacities; in the future, storage will be much more capacious and cheaper. Audio recording will be easy, and within a few years a video recording will be possible too.

I find the prospect of documented lives chilling, but some people will like the idea. For one thing, a documented life can be a good defense. We can think of the wallet PC as an alibi machine because encrypted digital signatures will guarantee an unforgeable alibi against false accusations. If someone ever accused you of something, you could retort: "Hey, buddy, I have a documented life. Those bits are stored away. I can play back anything I've ever said. So don't play games with me." Of course, if you were guilty of something, there would be a record of it. There would also be a record of any tampering you tried. Richard Nixon's taping of conversations in the White House—and then the suspicion that he tried to alter those tapes—contributed to his undoing. He chose to have a documented political life and came to regret it.

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Talks about what would become <EXIF data> in (pictures/videos/audio), and talks about how recording a bunch of shit would serve as <protection>.

Like the many times I talked about <Cortana> in (2020), how it would <protect the person carrying the device>. <Cortana> is just a <voice assistant>, but I was using it as a <codename> for something FAR more capable than that, something that would outperform a version of <ChatGPT> that doesn't even exist yet, on its' best day.

I threw out some wild ideas, but ultimately <the device would serve to protect the person carrying it>, maybe at some point in the future we develop <quantum tunneling

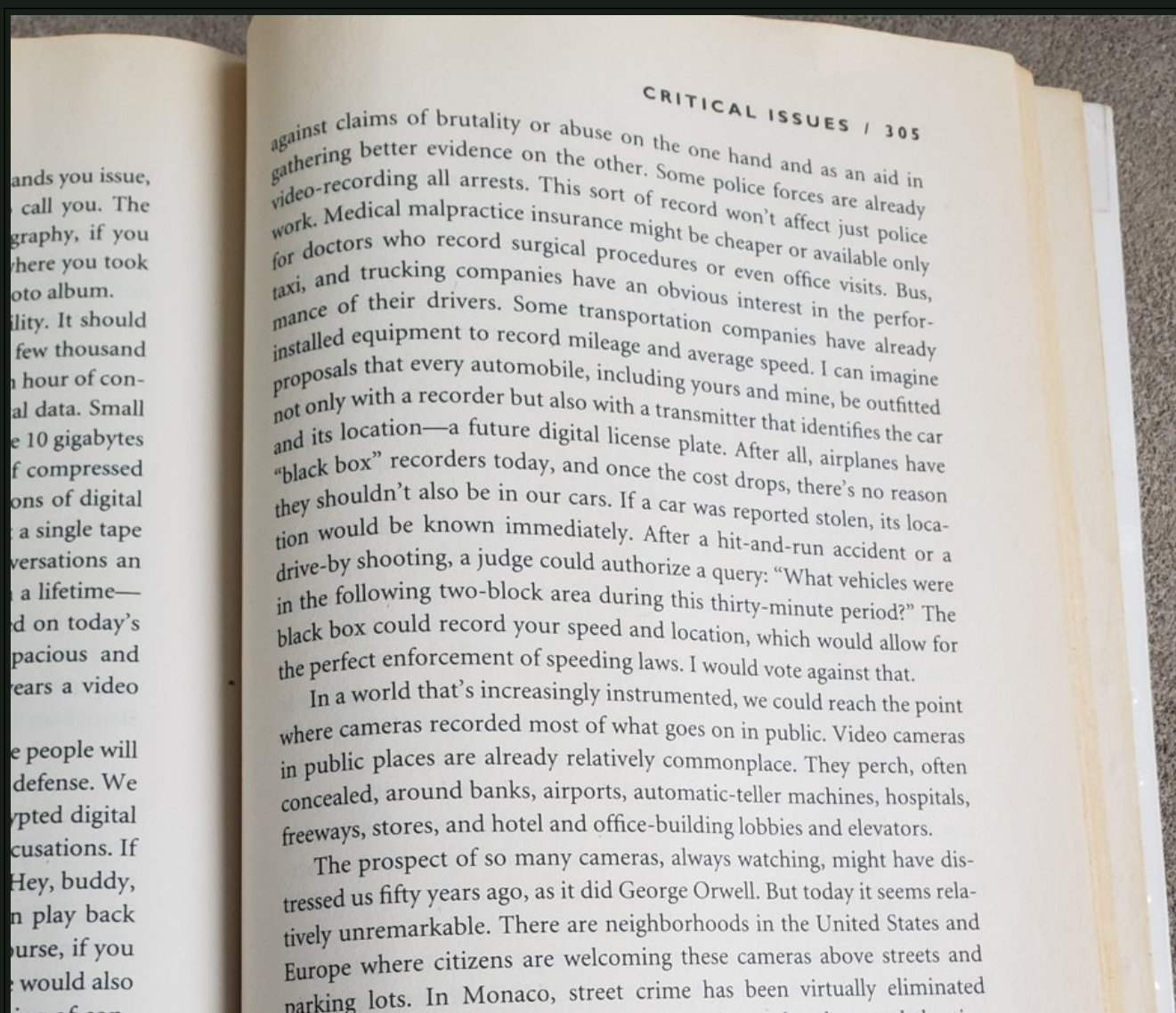
shields> that protect a person like a <forcefield>. Maybe that prediction is silly right NOW, but at some point it won't be.

Also predicts <body cameras used by police officers>.

Also talks about how <recording everything> is like <your life being an open book> ... Hey, buddy, I have a <documented life>. Those bits are <stored away>. I can <play back anything I've ever said>. So <don't play games with me>.

^ my exact mentality for years.

Hasn't ALWAYS been, but it has since I started my company.



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Talks about cheaper <medical malpractice insurance> offered to <surgeons> who <record their medical procedures/operations> on video, and the continued importance of <cameras/video>.

...can easily
imagine proposals that virtually every pole supporting a streetlight should also have one or more cameras. The images from these cameras might be accessed only in the event of a crime, and even then possibly only under court order. Some people might argue that every image from every camera should be available for viewing by everyone at any time. This raises serious privacy questions in my mind, but advocates might argue that it's appropriate if the cameras are only in public places.

Almost everyone is willing to accept some restrictions in exchange for a sense of security. It's a question of balance. From a historical perspective, people living in the Western democracies already enjoy a degree of privacy and personal freedom unprecedented in human history. If ubiquitous cameras tied into the network should prove to reduce serious crime dramatically in test communities, a real debate would begin over whether people fear surveillance more or less than they fear crime. It's difficult to imagine a government-sanctioned experiment along these lines in the United States because of the privacy issues it raises and the likelihood of constitutional challenges. However, opinion can change. It might take only a few more incidents within the borders of the United States like the bombings in Oklahoma City and Atlanta for attitudes toward strong privacy protection to shift. What today seems like digital Big Brother might one day become the norm if the alternative is being left at the mercy of terrorists and criminals. I'm not advocating a position, just pointing out that it's a political decision—and one over which there's likely to be a great deal of debate in the years ahead.

At the same time that technology is making it easier to create video

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Predicts <Flock> + <Palantir>-type companies in (1995/1996).

Predicts how <opinion> would <change> regarding <security and privacy> with <public facing cameras>, given an <incentive> like the bombings in <Oklahoma City> and <Atlanta>.

Predicted (9/11) to some extent? But not with any <clarity>.. Just that <big brother> would make themselves apparent with enough <terrorist pressure>.

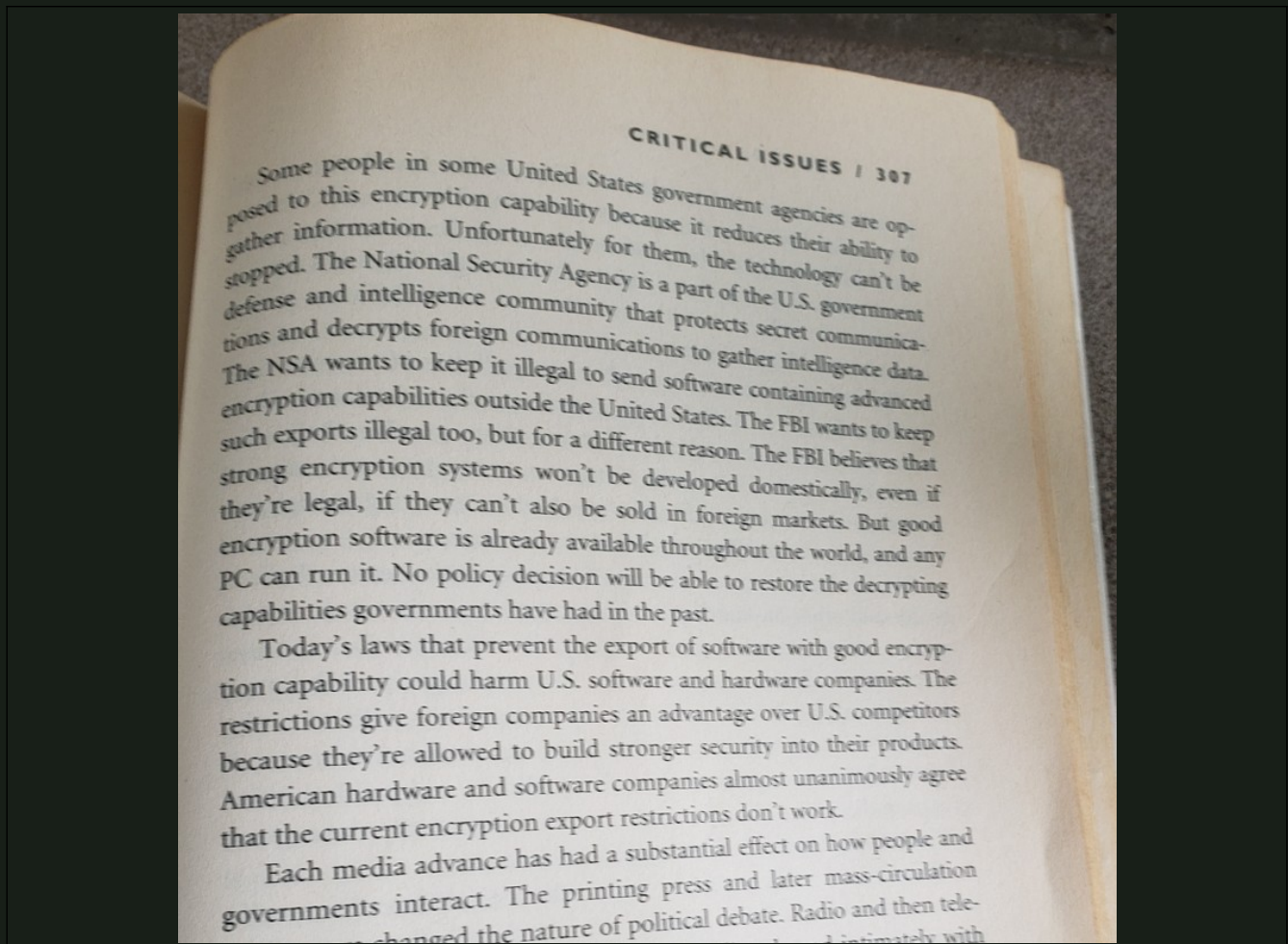
I.E. <Uniting and Strengthening America: Providing Appropriate Tools Required to Intercept and Obstruct Terrorism> (USA-PATRIOT Act), signed 10/26/2001.

The Act was passed extremely quickly — just 45 days after 9/11 — under high legislative urgency.

Event	Date
Windows XP RTM	Aug 24, 2001
Windows XP Retail	Oct 25, 2001
PATRIOT Act House Vote	Oct 24, 2001
PATRIOT Act Senate Vote	Oct 25, 2001
PATRIOT Act Signed	Oct 26, 2001

XP hit stores **the same day** the Senate passed the PATRIOT Act. Wild bit of historical overlap.

I believe there was **<another motivation>** for it being released on that **<exact day>**.



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Explains how the <National Security Agency> protects <secret communications> and <decrypts foreign communications> to <gather intelligence data>.

And, how the <FBI> wanted to keep such <exports illegal> too, but for a <different reason>. Key words → <government>, <decryption>.

What is it called, <Bull Run>?

[https://en.wikipedia.org/wiki/Bullrun_\(decryption_program\)](https://en.wikipedia.org/wiki/Bullrun_(decryption_program))

NSA tool that <decrypts> whatever (system/communications) on earth?

“Bullrun (BULLRUN) is a clandestine, <highly classified program> to <crack encryption> of <online communications + data>, which is run by the <United States National Security Agency> (NSA).”

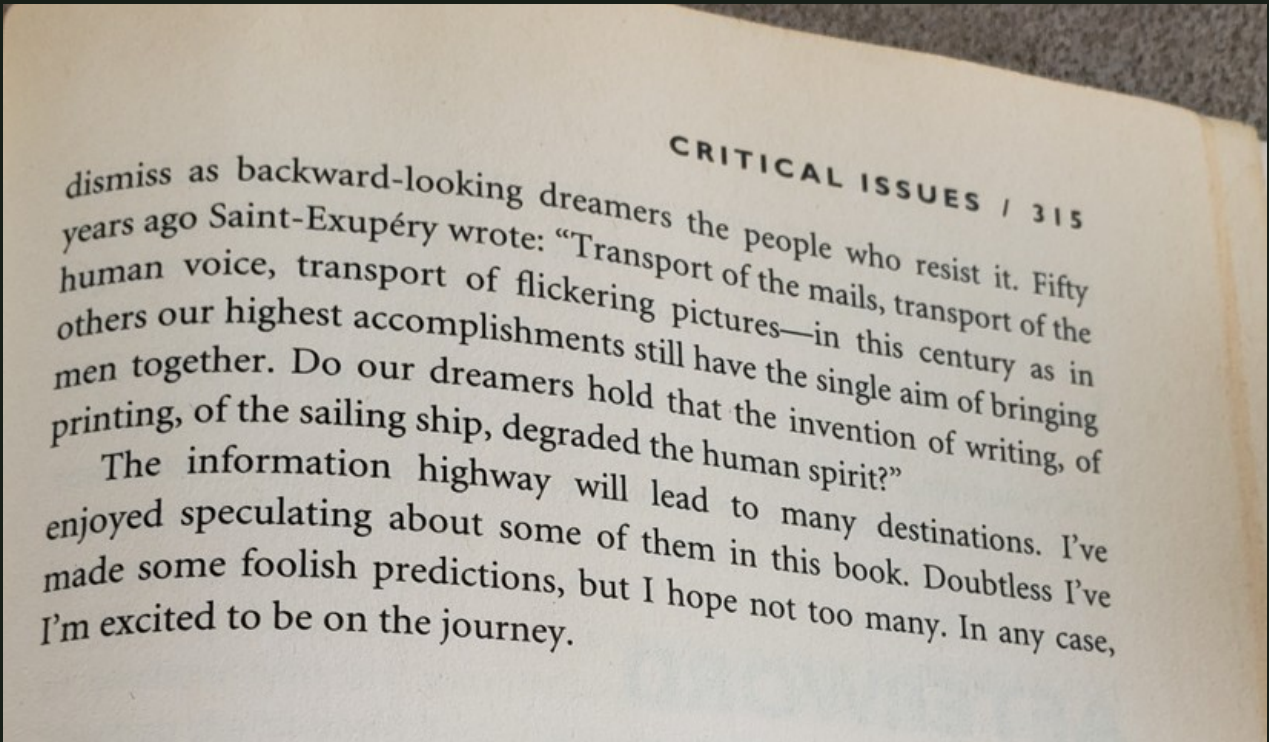
So, dude also predicted <Bullrun> (decryption program by the NSA), in (1995).

Named the 1) <specific agency>, and the 2) <specific task>.

ALMOST like the dude saw it coming~! No way~! (way, dude. way indeed.)

“Through the NSA-designed <Clipper chip>, which used the <Skipjack cipher> with an <intentional backdoor>, and using various specifically designed laws such as CALEA, CESA and restrictions on export of <encryption software> as evidenced by <Bernstein v. United States>, the <U.S. government> had <publicly attempted> in the (1990)’s to ensure its’ <access to communications> and <ability to decrypt>.

In particular, technical measures such as <key escrow>, a <euphemism for a backdoor>, have met with <criticism> and <little success>.”



dismiss as backward-looking dreamers the people who resist it. Fifty years ago Saint-Exupéry wrote: “Transport of the mails, transport of the human voice, transport of flickering pictures—in this century as in others our highest accomplishments still have the single aim of bringing men together. Do our dreamers hold that the invention of writing, of printing, of the sailing ship, degraded the human spirit?”

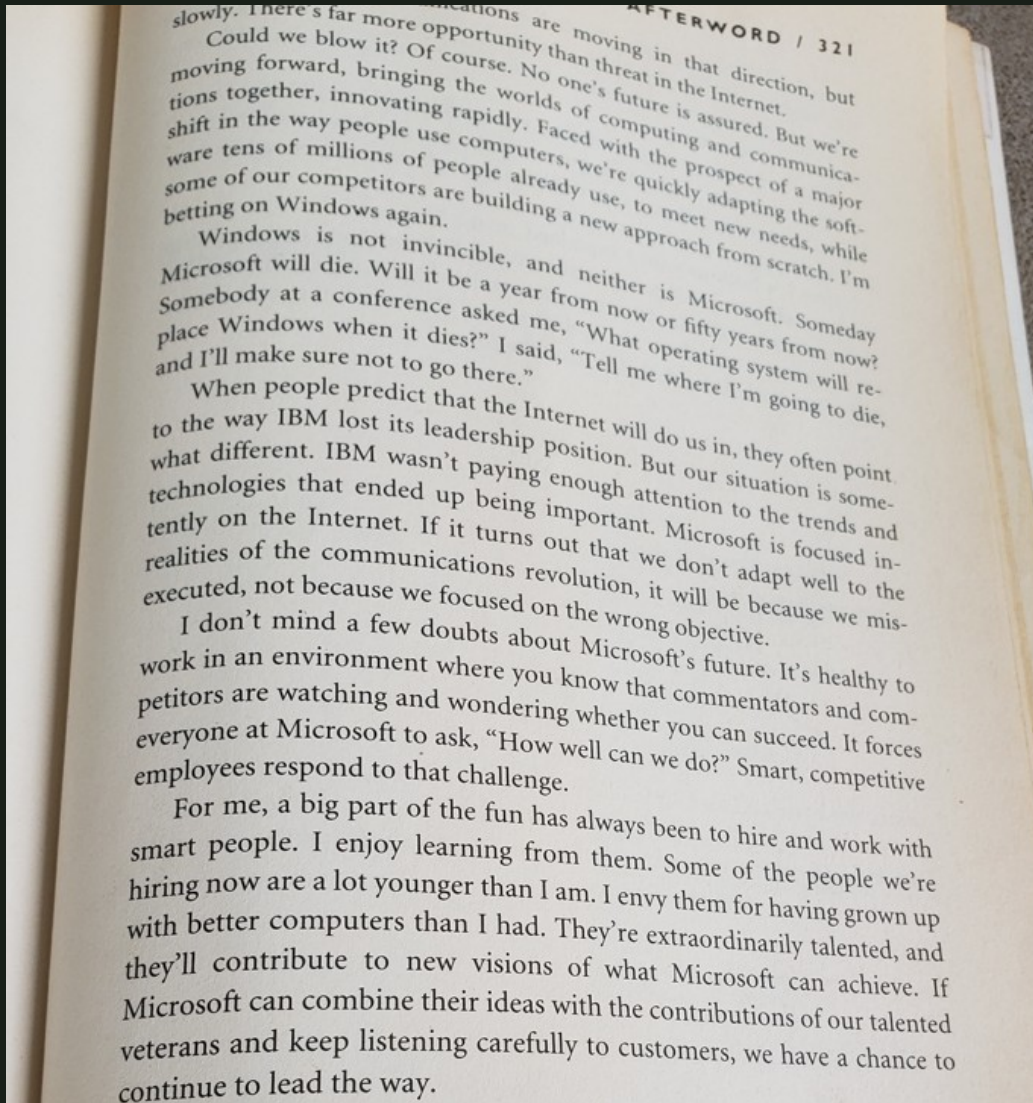
The information highway will lead to many destinations. I’ve enjoyed speculating about some of them in this book. Doubtless I’ve made some foolish predictions, but I hope not too many. In any case, I’m excited to be on the journey.

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Says "Doubtless, I've made some <foolish predictions>, but I hope not too many."

Comes off sounding like <Ken Jennings> saying "Look, I couldn't win Jeopardy if my life depended on it..." only to wind up becoming the <best contestant in history>.

Dude rattled off maybe 100+ <accurate predictions> that stood the test of time <throughout the entire book>, prompting me to <write about all of it>.



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Makes a humble claim that "Someday Microsoft will die", but I'm not seein' that happening anytime soon. They're here to stay... but like he says, <Windows/Microsoft> aren't invincible.

I think it just comes down to what he says "For me, a big part of the <FUN> has

always been to <hire + work with smart people>. I ENJOY learning from them. Some of the people were hiring now are a lot younger than I am. I envy them for having grown up with <better computers> than I had. They're <extraordinarily talented>, and they'll <contribute to new visions> of what <Microsoft> can achieve."

<Pretty humble statement>.

It's like hes saying "Look dude, I'm just a guy who had a dream and a vision, but I've only got so much left in the tank. Someone else will have to carry the torch."

So ultimately, <Microsoft's success> lay in the hands of the type of people he enjoyed working with, and learning from.

That's what I got out of this book, that it was the FUN he had that made it all come together. Ya know, if you're <not having fun> doing what you're doing, <what the hell are you doing it for>? Your best work comes out when you're having fun.

